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DATA ANALYSIS IN MINUTES, NOT MONTHS

RNA and small-RNA signatures in iPSC-derived microglia of Alzheimer's disease

An outlier-corrected reanalysis of GSE332551

Dataset	GEO GSE332551
Source study	Wu, Choi, Liu et al., <i>Frontiers in Neuroscience</i> (2026)
Design	3 sporadic AD vs 3 cognitively normal iMG (APOE3/3)
Assays	Total RNA-seq, T4 PNK small-RNA-seq
Method	DESeq2 v1.46, R17_AD outlier excluded
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Date	July 2026

Verdict. The biology holds up. The small-RNA class shifts and the marker-gene directions both reproduce from the public data. The exact RNA-seq DEG count does not, because it is unstable at $n = 3$ and was inflated by one low-depth outlier (R17_AD). This report works from an outlier-corrected, FDR-based DEG list. Beyond reproduction, threshold-free pathway and regulator analyses point to an inflammatory, lipid-laden microglial state driven by SREBP.

Abstract

Background. Alzheimer's disease (AD) reshapes microglial gene and small-RNA programs, but patient-derived models are scarce. The original study profiled iPSC-derived microglia-like cells (iMG) from 3 sporadic AD donors and 3 cognitively normal (CN) controls, matched for age and sex and all APOE3/3, using total RNA-seq and T4 PNK small-RNA-seq. We reanalyzed the deposited count matrices to check whether the reported differential expression reproduces from public data and to put the DEG calls on a firmer footing.

Methods. We ran DESeq2 (v1.46) on both matrices (AD vs CN, $n = 3$ per group). RNA-seq transcript counts were collapsed to genes, leaving 13,611 genes after a mean-count filter of 10. Small-RNA features were tested directly (1,467 kept at mean ≥ 5) and classified as piRNA, tRF, miRNA, or snoRNA. We added normalization checks (library size, DESeq2 size factors, inter-sample correlation, PCA), applied log-fold-change shrinkage with FDR control, and re-ran the RNA-seq analysis with the outlier removed. For interpretation we ran preranked GSEA (Wald-ranked; Hallmark, GO-BP, Reactome, KEGG), ssGSEA microglial-state scoring, transcription-factor activity inference (decoupler ULM against the signed CollecTRI regulons), and a miRNA-target integration against validated miRTarBase targets.

Results. The small-RNA results reproduced well: 68 differential sncRNAs (the paper reported 64), a fold-change correlation of $r = 1.000$, and the same class pattern (piRNAs and tRFs up in AD, miRNAs and snoRNAs mostly down). The RNA-seq results did not reproduce as cleanly. The deposited matrix gave 827 DEGs with a 3.6-to-1 skew toward "up in AD", and that skew traced back to one sample, R17_AD. Dropping it removed the skew (387 DEGs, 134 up and 253 down), raised the FDR-significant genes from 11 to 30, and left 13 high-confidence DEGs. Every named marker gene from the paper came back in the right direction (11 of 11 up, 18 of 18 down).

Downstream interpretation. Threshold-free GSEA (555 sets at $FDR < 0.25$) overlaps the paper on the translation/ribosome down-signal but adds up-signals it did not emphasize: neuroinflammation (TNF/NF- κ B) and cholesterol/lipid biosynthesis. ssGSEA scores the AD cells toward inflammatory, cholesterol/SREBP and lipid-droplet (LDAM) states, and TF-activity inference attributes the lipid arm to significant SREBF1 ($FDR = 0.004$) and SREBF2 ($FDR = 0.02$) activation while independently recovering MHC class II (RFX complex) repression, matching the paper's HLA-DR observation. Across the two assays, the targets of the down-in-AD miRNAs are coherently de-repressed in the transcriptome (pooled $p = 0.015$).

Conclusions. Work from an outlier-corrected, FDR-based DEG list, and lean on the signature-level read-outs (GSEA, state scoring, TF activity), which stay robust at this sample size; together they describe an inflammatory, lipid-laden microglial state driven by SREBP. All results, figures, and scripts accompany this report.

1 Graphical abstract

The whole project on one figure: the study inputs, the reanalysis steps, and what came out of each assay.

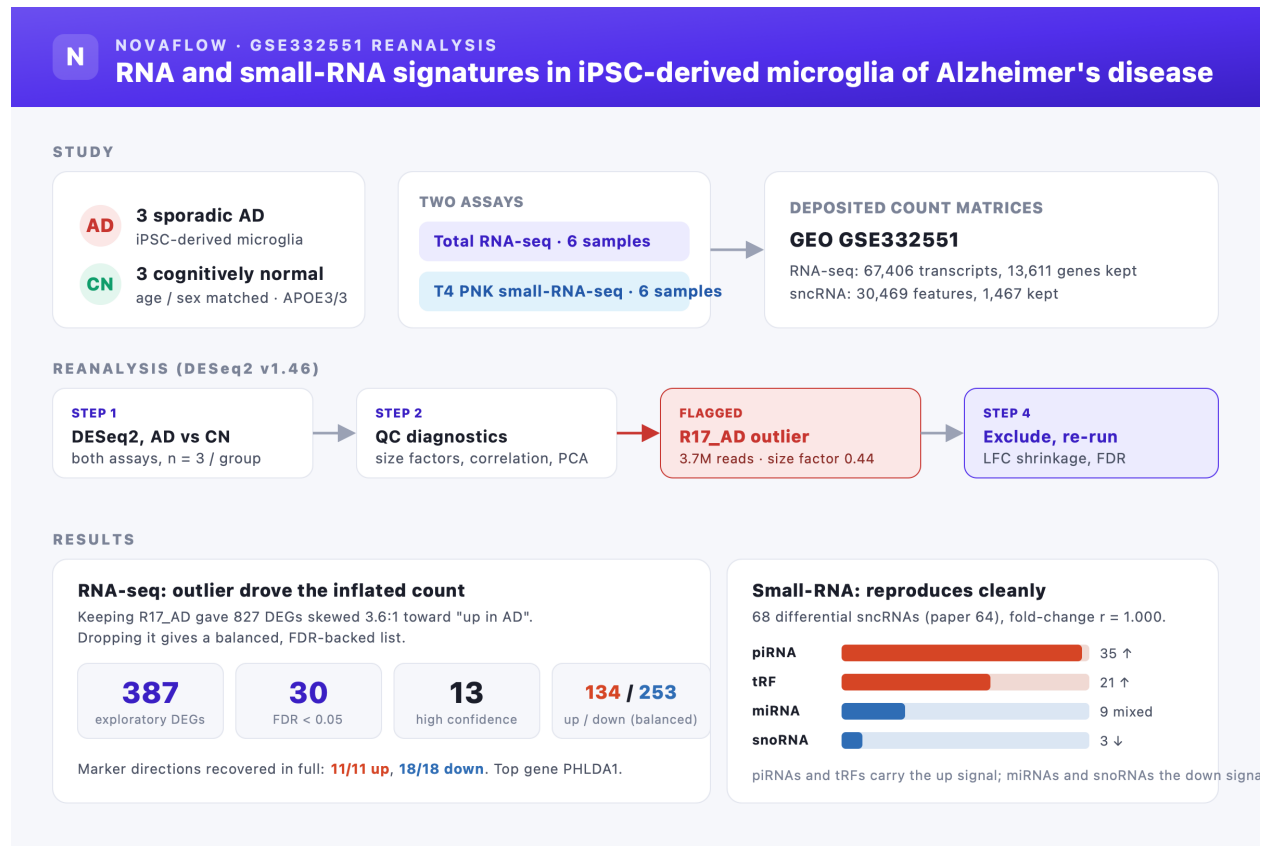


Figure 1: Study design, DESeq2 reanalysis, the R17_AD outlier that inflated the RNA-seq count, and the corrected RNA-seq and small-RNA outputs.

2 Study design and methods

- **Samples.** iPSC-derived microglia-like cells from 3 sporadic AD patients and 3 cognitively normal controls, matched for age and sex, all APOE3/3. Six RNA-seq and six small-RNA-seq libraries.
- **RNA-seq.** 67,406 transcripts collapsed to genes; 13,611 genes retained at mean count ≥ 10 . Tested AD vs CN with DESeq2, paper-like thresholds $p < 0.05$ and $|\log_2 FC| > 0.8$.
- **Small-RNA-seq.** 30,469 features; 1,467 retained at mean ≥ 5 . Tested directly with DESeq2 ($|\log_2 FC| > 1$) and classified into piRNA, tRF, miRNA, snoRNA.
- **Corrections.** Normalization diagnostics (library size, size factors, correlation, PCA), log-fold-change shrinkage, FDR control, and exclusion of the R17_AD outlier.

3 Reproduction and the R17_AD outlier

Applying paper-like thresholds to the deposited RNA-seq matrix produced far more DEGs than the paper reported, with a strong skew toward "up in AD". The cause was one sample, R17_AD. It has about half the library depth of any other sample and a size factor of 0.44, it correlates only 0.77–0.82 with the others (where they correlate 0.92–0.96 among themselves), and it sits alone on PCA. With only three samples per group, normalization scales its low counts up and manufactures "up in AD" genes, which is the source of the 3.6-to-1 skew.

Table 1: Per-sample QC. R17_AD stands apart on every measure.

Sample	Library size	Size factor	Median Spearman
R15_AD	9.1M	1.05	~0.95
R17_AD	3.7M	0.44	~0.80
R18_AD	8.5M	0.97	~0.95
R19_CN	8.4M	1.00	~0.92
R20_CN	15.2M	1.76	~0.95
R22_CN	12.0M	1.35	~0.95

Table 2: DEG counts across analyses. Excluding R17_AD restores a balanced list and triples the FDR-significant genes.

Analysis	Total	Up in AD	Down in AD	Interpretation
Paper RNA-seq	195	100	95	Published nominal list
Original gene-level	827	649	178	Inflated, skewed by R17_AD
Transcript-level	1382	1038	344	Same skew, not gene collapsing
Corrected exploratory	387	134	253	Balanced; raw-p tier
Corrected high-confidence	13	6	7	FDR + shrunken FC tier

Marker-gene *directions* are fully recovered on the gene-level shrunken analysis: 11 of 11 paper “up in AD” genes and 18 of 18 “down in AD” genes test in the correct direction.

4 Corrected RNA-seq results

After excluding R17_AD, 13 genes pass FDR < 0.05 with a meaningful shrunken fold-change.

Table 3: The 13 high-confidence DEGs from the corrected analysis.

Gene	log2FC raw	log2FC shrunken	FDR	Direction
PHLDA1	2.059	1.495	4.5e-11	up in AD
SLC44A2	-1.363	-0.988	6.6e-04	down in AD
ALK	2.794	1.053	1.3e-03	up in AD
KCNQ3	-1.156	-0.885	1.8e-03	down in AD
LRRC4	-1.318	-0.927	2.6e-03	down in AD
PLOD2	-2.552	-0.840	8.1e-03	down in AD
CKB	-2.185	-0.873	8.3e-03	down in AD
ZFP3	-2.292	-0.865	8.3e-03	down in AD
LRP8	1.179	0.837	1.5e-02	up in AD
RYR1	-1.712	-0.869	1.8e-02	down in AD
AP1S2	1.406	0.868	2.5e-02	up in AD
MID1IP1	1.270	0.825	3.2e-02	up in AD
SLIT3	1.378	0.837	3.4e-02	up in AD

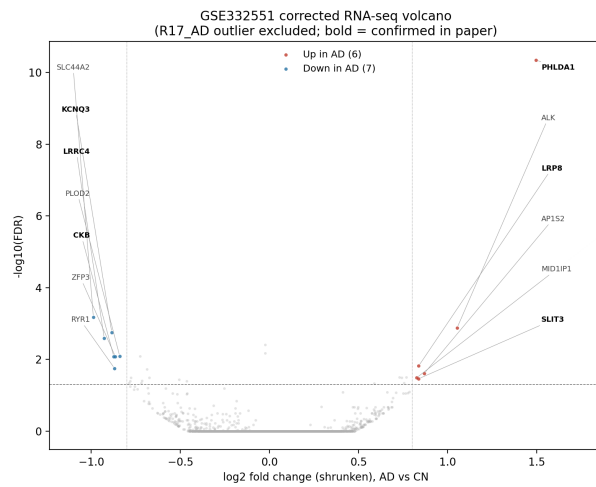


Figure 2: Corrected volcano. Shrunken $\log_2$ fold-change against FDR, R17_AD excluded.

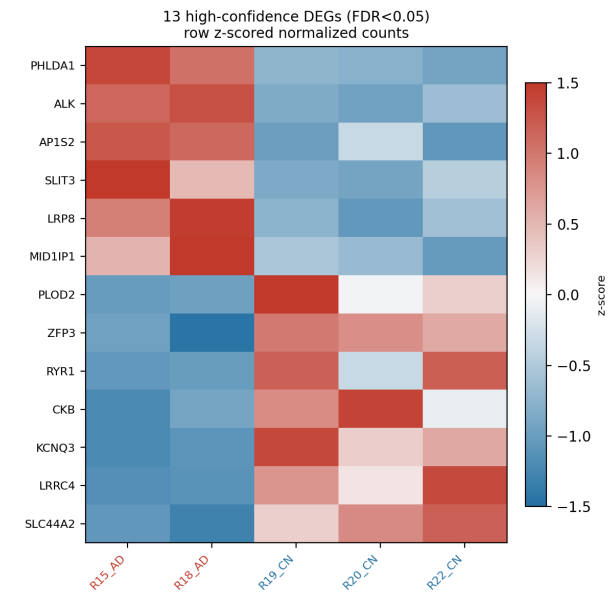


Figure 3: High-confidence DEG heatmap. Row z-scored normalized counts across the six samples.

5 Small-RNA results

The small-RNA analysis reproduces well: 68 differential sncRNAs (paper 64), fold-change $r = 1.000$, with the class pattern the paper described. piRNAs and tRFs carry the up-in-AD signal; miRNAs and snoRNAs carry the down signal.

Table 4: Differential small-RNA by class (reanalysis).

Class	Up in AD	Down in AD	Total
piRNA	34	1	35 (51%)
tRF	21	0	21 (31%)
miRNA	4	5	9 (13%)
snoRNA	0	3	3 (4%)

Notable sncRNAs. tRF5-Lys-CTT-6 is the highest-count up tRF ($\log_2FC \approx +3.5$); hsa-piR-23444 is the most up piRNA ($\approx +4.3$); SNORD123 is the most down snoRNA (≈ -5.1); miR-143-3p, miR-145-5p, and miR-181a-3p are AD-associated miRNAs that go down.

6 Pathway analysis and cross-assay integration

Because a DEG list is unstable at $n = 3$, we ranked all 13,523 genes by their Wald statistic and ran preranked GSEA against Hallmark, GO-BP, Reactome and KEGG. This is threshold-free, so it does not depend on any cutoff. 555 gene sets are enriched at $FDR < 0.25$ (377 up, 178 down). They overlap the paper on the translation/ribosome down-signal but surface inflammation and cholesterol/lipid programs it did not emphasize (the paper's headline chromatin/exosome GO terms are not among our

top hits), all without relying on the fragile count. The up-in-AD side is dominated by **neuroinflammation** (TNF- α /NF- κ B, inflammatory response, IL-10) and **microglial lipid metabolism** (cholesterol and sterol biosynthesis, PPAR, mTORC1); the down-in-AD side is a coordinated collapse of the **translation / ribosome** program (NES down to -3.0).

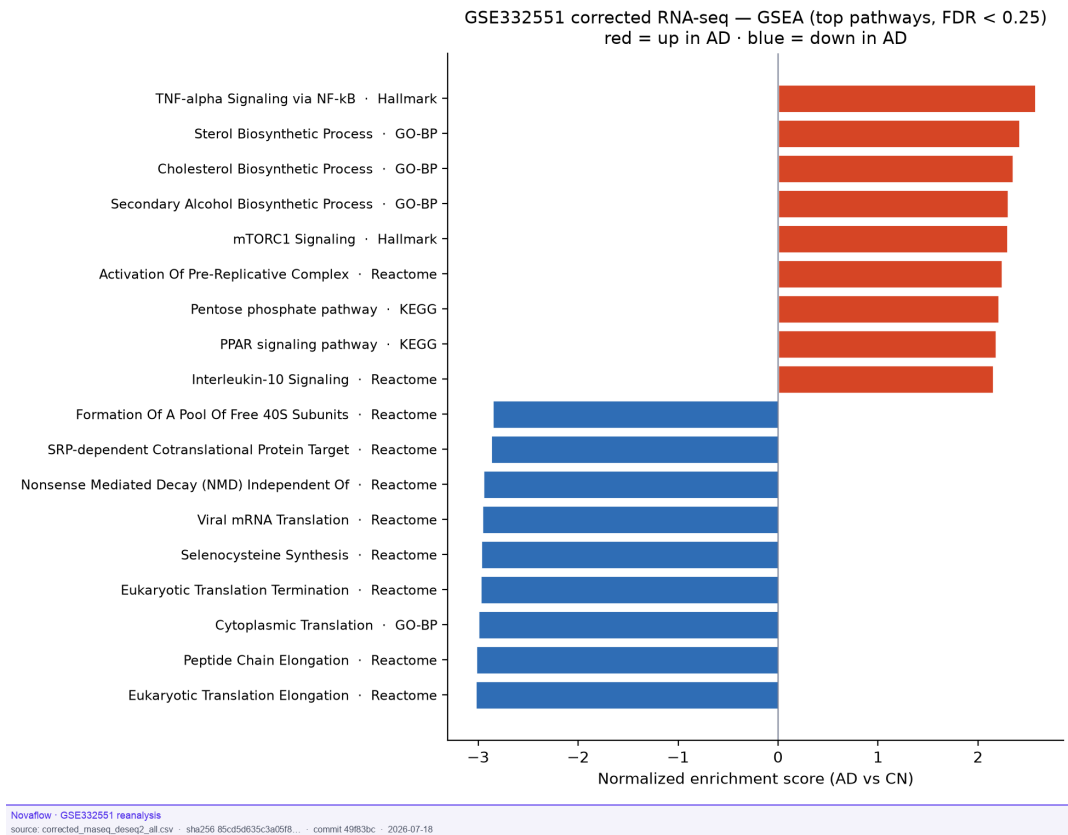


Figure 4: Top GSEA pathways for the corrected RNA-seq analysis (FDR < 0.25). Red = up in AD, blue = down in AD.

We then ran the one genuinely cross-assay test the deposited data supports. If the differential miRNAs are functional, their targets should move the opposite way in the RNA-seq. Comparing each miRNA's validated targets (miRTarBase) against the genome background, the **targets of the five down-in-AD miRNAs shift up** in the transcriptome (pooled median $+0.13$, one-sided $p = 0.015$; all five concordant), with miR-224-3p ($p = 0.027$) and the paper's miR-145-5p ($p = 0.043$) individually significant. The up-in-AD miRNAs do not show clean target repression (pooled $p = 1.0$); three of four are broad let-7 / miR-20b family members with 600–1100 targets each, so the effect is diluted and, at $n = 3$, underpowered.

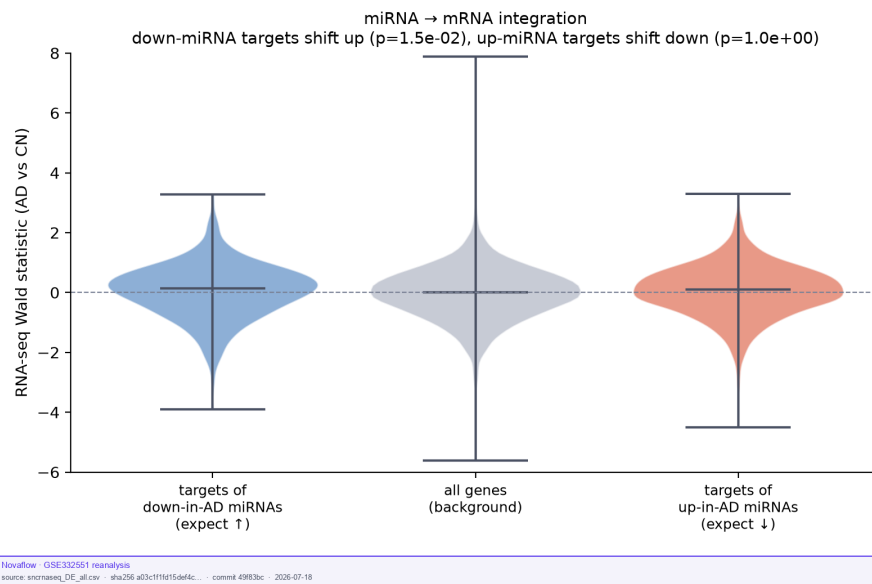


Figure 5: miRNA → mRNA integration. RNA-seq Wald statistic for targets of down- and up-in-AD miRNAs vs all genes. Down-miRNA targets are de-repressed; up-miRNA targets are not repressed.

7 Microglial state and upstream regulators

The pathway results say what changed; two signature-level analyses say which microglial state the cells moved toward and which regulators drive it (both far more stable at $n = 5$ than per-gene tests). ssGSEA scoring against defined programs shows the AD iMG shift toward **inflammatory (NF- κ B)**, **cholesterol/SREBP** and **lipid-droplet (LDAM)** states with **translation down** — the disease-associated, lipid-laden direction.

TF-activity inference (decoupler ULM against the signed CollecTRI regulons, on the AD-vs-CN Wald contrast) then pins the lipid arm on a statistically significant activation of **SREBF1** (activity +5.8, FDR = 0.004) and **SREBF2** (+5.2, FDR = 0.02), the master regulators of cholesterol biosynthesis. NF- κ B, IRF1 and PPAR δ are directionally activated but underpowered at this n . Independently, the **RFX complex** (RFXANK, FDR = 0.02; RFXAP, RFX5) — which drives MHC class II — is significantly *repressed*, recovering the paper’s reported HLA-DR reduction directly from the transcriptome.

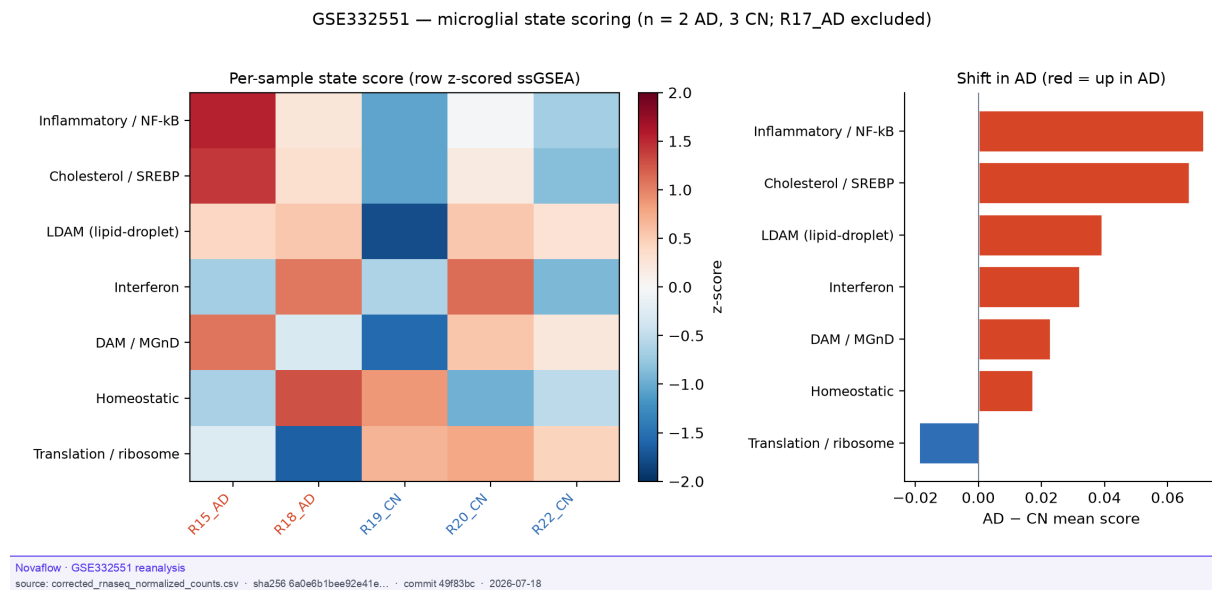


Figure 6: Microglial state scoring (ssGSEA). Left: per-sample row z-scored scores (AD red, CN blue). Right: AD – CN shift.

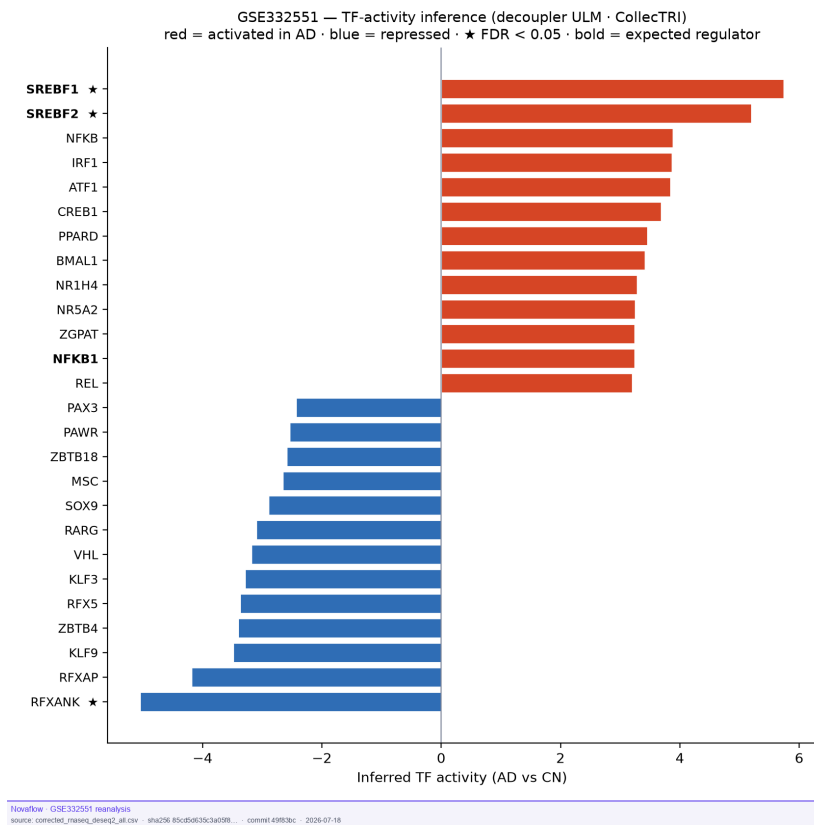


Figure 7: Inferred TF activity (decoupler · CollecTRI). SREBF1/2 activated (FDR-significant); RFX complex repressed. ★ FDR < 0.05.

8 Conclusions

- The underlying biology reproduces: small-RNA class shifts and every named marker-gene direction hold up.
- The exact RNA-seq DEG count does not reproduce, because it is unstable at $n = 3$ and was inflated by one low-depth outlier (R17_AD).
- Downstream work should use the outlier-corrected, FDR-based DEG list (30 genes at $FDR < 0.05$; 13 high-confidence).
- Threshold-free GSEA is the more robust read-out at $n = 3$: inflammation and cholesterol/lipid metabolism up, translation/ribosome down. The down-in-AD miRNAs' targets are coherently de-repressed in the transcriptome, tying the two assays together.

Analysis scripts (run in order): 01_deseq2_rnaseq.R, 02_deseq2_sncrnaseq.R, 03_visualize.py, 04_paper_comparison.py, 05_reproduce_paper.R, 06_diagnose_normalization.R, 07_corrected_DEGs.R, 08_visualize_corrected.py. Full source and an interactive DEG table are in the accompanying portal.